

```

algo(A, l, h) /* l=0, h=N */
{
  for(i=l; i<h; ++i)
    istruzione /* costo costante */

```

```

  if (h-l > 1) then

```

```

  {
    c = Math.random() /* genera un numero casuale
                       in tempo costante */

```

```

    if (c > 0.5) then

```

```

      algo(A, l, l + (h-l)/2) ←

```

```

    else

```

```

      { algo(A, l, l + (h-l)/2) ←

```

```

        algo(A, l + (h-l)/2, h) ←

```

```

      }
    }
  }

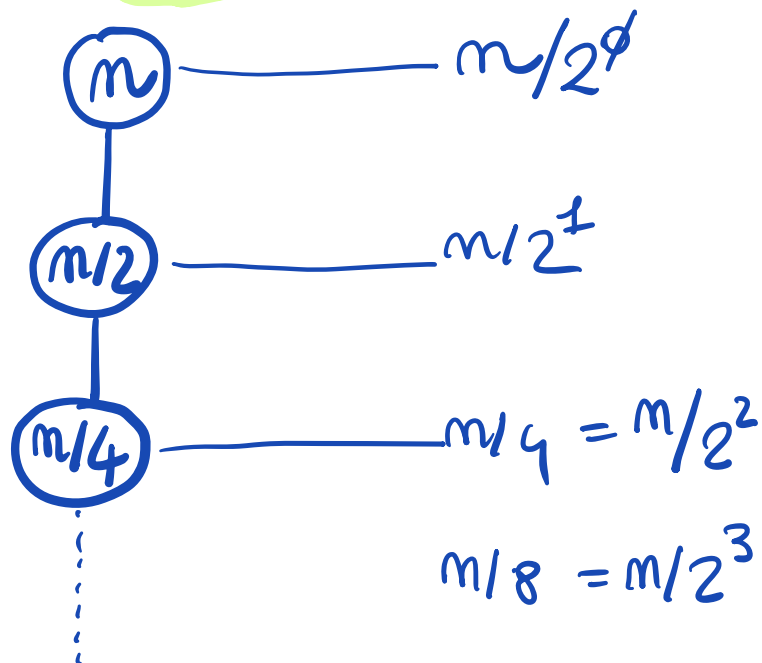
```

$$T(n) = 1 \cdot T(n/2) + n \quad \text{MIGLIORE}$$

$$T(n) = 2 \cdot T(n/2) + n \quad \text{PEGGIORE}$$

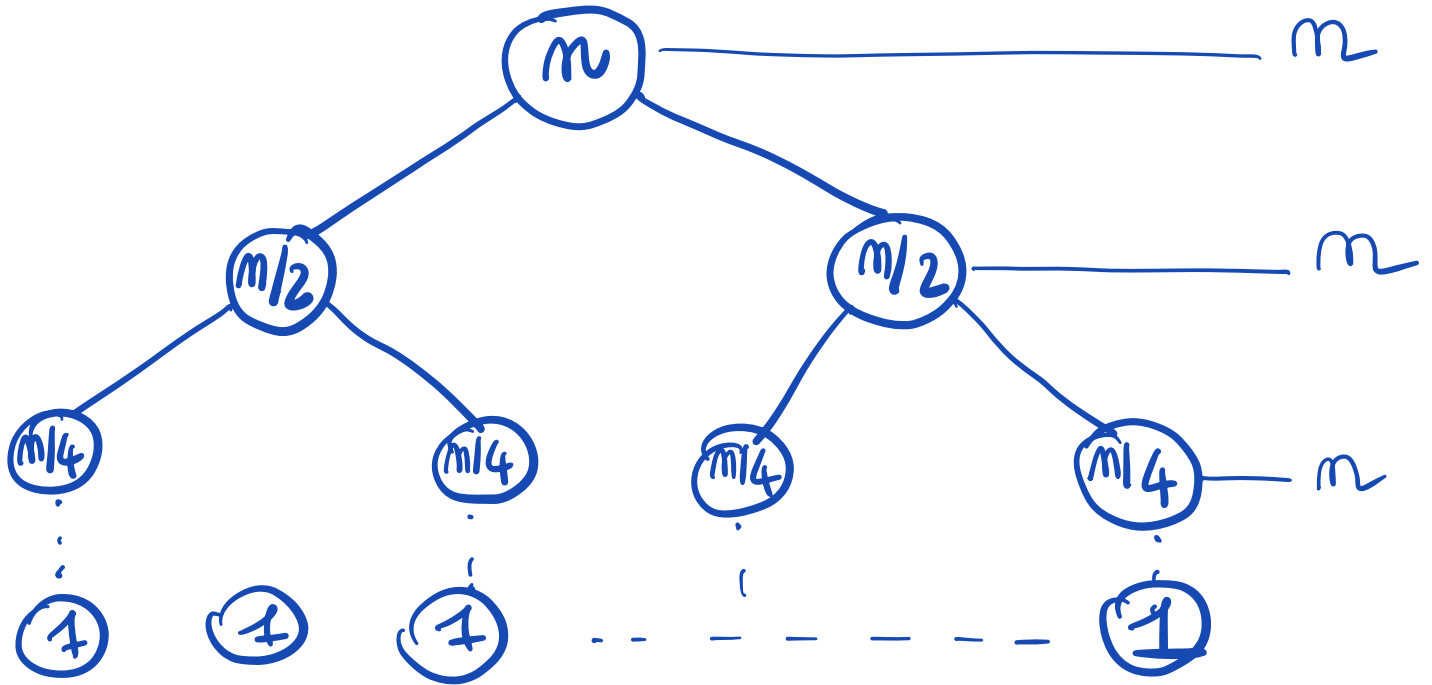
$$\sum_{i=0}^{\log n} \frac{n}{2^i}$$

$$n \cdot \sum_{i=0}^{\log n} \left(\frac{1}{2}\right)^i$$



$$\rightarrow \textcircled{1} \text{ ————— } 1 \quad \frac{n}{2^k} = 1 \Rightarrow k = \log n$$

$$\boxed{O(n)}$$



$$\frac{n}{2^k} = 1 \Rightarrow k = \log n$$

$$\rightarrow \sum_{i=0}^{\log n} n = \underline{O(n \cdot \log n)}$$

Algo(m)

```

{ a = ∅; j = 1; k = ∅
  while j < m
  {
    k++
    for (i = 1; i ≤ k; i++)
    {
      h = 2
      while h < 2m
      {
        a++
        h = h · h
      }
      j = 2 · j
    }
  }

```

log m

1 + 2 + 3 +
4 + ... log m

$2^0 \quad 2^1 \quad 2^2 \quad 2^3 \quad 2^4$
j = 1; 2; 4; 8; 16

$\log_2(2^x) < \log_2(m)$

$h = 2; 4; 16; 16 \cdot 16; \dots$
 $2; 2^2; 2^4; \dots$

$\log(2^{2^x}) < \log(2^m)$

$\log(2^x) < \log(m)$

$x < \log m$

$$\left(\sum_{k=1}^{\log(m)} k \right) = \frac{\log m (\log m + 1)}{2} \approx \log^2 m$$

$$\rightarrow \left(\begin{array}{c} 1 \\ \log m \end{array} \right) + \left(\begin{array}{c} 2 \\ \log m - 1 \end{array} \right) + \left(\begin{array}{c} 3 \\ \log m - 2 \end{array} \right) + \left(\begin{array}{c} 4 \\ \log m - 3 \end{array} \right) + \left(\begin{array}{c} 5 \\ \log m - 4 \end{array} \right) + \left(\begin{array}{c} 6 \\ \log m - 5 \end{array} \right) + \dots + \left(\begin{array}{c} \log m \\ 1 \end{array} \right)$$

$$O(\log^3 m)$$

$O(m)$

$\text{for } (i=m; i>0; i--)$ $m \cdot \log m \cdot J$

$O(\log m)$

$\text{for } (j=1; j<m; j=2 \cdot j)$

$\rightarrow \text{for } (k=0, \boxed{k < j}, k++) \leftarrow$

*# operazioni di costo costante **

$1 + 2 + 4 + \dots + 2^{m-1} \approx 2^m \approx m$

2^x

$j = 1; 2; 4; 8; 16; \dots$
 $2^0 \quad 2^1 \quad 2^2 \quad 2^3 \quad 2^4 \quad \dots$
 $\boxed{2^x} < m \Rightarrow < \log m$

$O(m^2) *$

$j = 1$

$1 + 2 + 4 + 8 + \dots$

$\text{for } (i=m, i>0, i--)$

$\text{for } (j=1, j<m, j=2 \cdot j) \leftarrow$

$\text{istruzione di costo } m *$

$j = 1 \quad j = 2 \quad j = 4 \quad j = 8 \quad j = 16$

$$\sum_{x=1}^{\log m} \underbrace{m}_{\substack{m \\ + \\ m \\ + \\ m \\ + \\ m \\ + \\ m}} = \boxed{m \cdot \log m}$$

for (j=1; j<m; j=2*j)
 → for (k=0; k<j; ++k)
 istruzione di costo j

$$\begin{array}{cccccc} j=1 & j=2 & j=4 & j=8 & j=16 & \\ \boxed{1} + \boxed{2} + \boxed{4} + \boxed{8} + \boxed{16} + & & & & & \\ & & & \approx 2^m \approx m & & \end{array}$$

$$\sum_{x=1}^{\log m} 2^x$$

$$i = m$$

$\log m$ — while $i > 1$
 $\{$ $j = 1$ —

$\log m$ — while $j < m$
 $\{$ $k = \emptyset$

$\text{while } k < m$
 $k = k + 2$

 $j = 2 \cdot j$
 $\}$
 $i = i / 2$
 $\}$

$$\frac{m}{2}$$

$$m, \frac{m}{2}, \frac{m}{4}, \frac{m}{8}, \dots$$

$$\frac{m}{2^x} = 1$$

$$x = \log m$$

$$2^y = m$$

$$\Rightarrow y = \log m$$

$$0 \quad 2 \quad 4 \quad 6 \quad 8 \quad \dots \quad m$$

$$O(\log m \cdot \log m \cdot \frac{m}{2})$$

$$O(m \cdot \log^2 m)$$

$$T(m) = a T(m/b) + f(m)$$

$$T(m) = a T(m/b) + f(m)$$

$$f(m) \sim m^{\log_a b}$$

$$a > 1$$

$$b > 1$$

$$m$$

$$m^{\log_a 1} = 1$$

CASE 3

$$f(m) = \Omega(m^{\log_a b + \epsilon})$$

$$m^{\boxed{1}} = \Omega(m^{\phi + \epsilon}) = \Omega(m^{\boxed{\phi + 0.1}})$$

$$\underline{a} f\left(\frac{n}{b}\right) \leq \underline{c} \cdot f(n) \quad \Leftarrow$$

$$\rightarrow \frac{n}{10/9} \leq c \cdot n = \frac{9}{10} \cdot n \leq \underline{c} \cdot n$$

$$c = \frac{9}{10}$$

$$\Theta(f(n))$$

$$\Theta(n)$$